

Battery activity driven representative day selection to accelerate degradation estimation process

Shah Mohammad Mominul Islam
Adelaide University
& Imperial Engineering Education
Adelaide, Australia
shah.islam,

S. Ali Pourmousavi
Adelaide University
Adelaide, Australia
a.pourm,

Wen L. Soong
Adelaide University
Adelaide, Australia
wen.soong@adelaide.edu.au,

Hossein Ranjbar
OptiGrid Pty Ltd
Adelaide, Australia
hossein@optigrid.energy

Abstract—Representative-day methods are widely used to reduce the computational burden of long-term battery degradation analysis in planning and design studies. However, conventional selection methods for representative days, such as cluster-mean or drought-day selection, are not always suitable for degradation analysis because they may not identify the battery activity patterns that drive ageing. This paper proposes two representative-day selection methods for the existing degradation estimation framework: 1) balanced positive and negative area (Method-I) and 2) frequent zero crossings (Method-II). The proposed selection methods are tested over ten years of data and compared against the Base Case and the existing cluster-mean selection method. The results show that both proposed methods improve the accuracy of the annual degradation estimation, with average relative errors reduced from 21.83% for the cluster-mean method to 5.12% and 6.71% for Method-I and Method-II, respectively. Among the two methods, Method-I gives the best overall performance.

Index Terms—Battery degradation, representative days, renewable microgrid, net demand.

I. INTRODUCTION

Lithium-ion batteries are widely used in renewable microgrids (MGs) because they improve reliability and provide operational flexibility. However, battery degradation cannot be ignored in planning and sizing studies because it affects usable capacity, replacement timing, and long-term system cost [1]–[6]. For this reason, many recent studies have included the cost of battery degradation in MG planning, scheduling, and operation analysis [1]–[5], [7], [8]. A major challenge is that an accurate degradation analysis usually requires high-resolution battery operation over long time horizons. A complete day-by-day simulation over many years until battery end-of-life (EoL) can provide a reliable benchmark, but it is computationally expensive and difficult to use in optimisation studies [3], [6]. To reduce this burden, representative-day and time-series aggregation methods are often used to compress the annual dataset while preserving the main operating patterns [9]–[14]. These methods are attractive because they can greatly reduce the simulation time. However, most existing representative-day methods are designed to match the overall annual profile

for battery sizing purposes, rather than the detailed battery activity within each day. Therefore, they may miss the charge-discharge patterns and switching behaviour that are more directly related to battery degradation. Therefore, there is still a need for representative-day selection methods that are fast but also more suitable for estimating battery degradation. Conventional representative-day selection methods are often based on general profile similarity, such as statistical grouping, average daily shape, cluster centroids, or renewable drought conditions [9]–[14]. For battery degradation analysis, the worst day is not necessarily the day with no solar and wind generation (known as drought days), typically these days do not have any battery activity. It is also not always the cluster mean net demand (P_{net}) day. A very drought day may keep the battery in a long discharge state with little chance to charge, while a mean day may smooth out important short-term variations. In contrast, battery degradation is more strongly affected by battery activity intensity, which depends on the temporal behaviour of P_{net} over the day, rather than only its average level. Representative-day selection for degradation studies should consider the shape and switching behaviour of P_{net} more carefully. This is because a drought day mainly represents a long energy shortage condition, but battery degradation is more related to how often and how strongly the battery moves between charging and discharging. Similarly, a similarity-based mean day can hide short-term peaks, valleys, and sign changes in P_{net} , so it may not preserve the battery cycling behaviour that is important for accurate degradation estimation.

The main hypothesis of this work is that a representative day for battery degradation studies should preserve the net-demand behaviour that leads to high battery usage. In particular, a day with large positive and negative P_{net} areas is expected to represent a day with high charge and discharge energy throughput. This is important because higher battery throughput usually means more battery cycling, and more cycling generally leads to higher degradation. Similarly, a day with frequent zero crossings in P_{net} is expected to represent repeated changes between charging and discharging conditions. Such frequent switching can indicate more cycle events within the day, which is also important for degradation estimation. Based on this idea, representative days selected using these two P_{net} characteristics may provide more accurate battery degradation estimates than conventional methods such as cluster-mean or drought-day selection.

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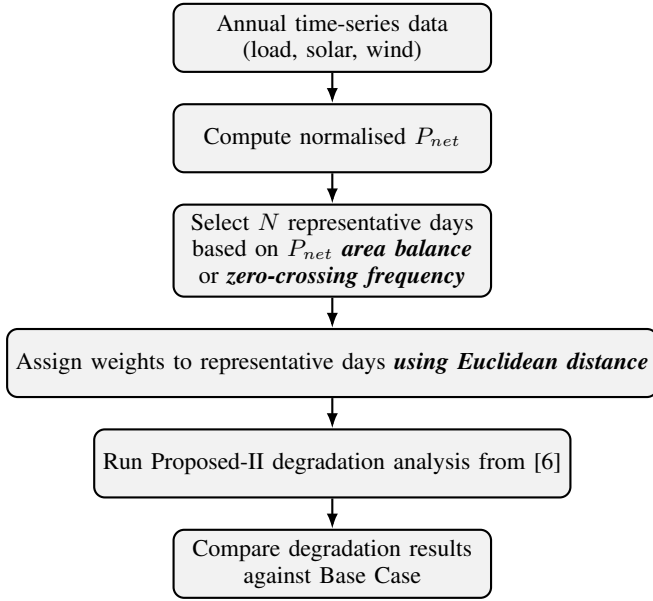


Fig. 1: High-level flowchart of the proposed representative-day-based degradation analysis framework.

The main contributions of this study are as follows:

- To propose two P_{net} -driven representative-day selection methods for battery degradation-focused studies.
- To embed both representative-day selection methods into the Proposed-II battery degradation framework of [6].

The rest of this paper is organised as follows: Section II presents the methodology, including the high-level workflow, the benchmark representative-day method, the two proposed P_{net} -based representative-day selection methods, the weighting process and the way these methods are embedded into the Proposed-II degradation framework of [6]. Section III presents and discusses the results, including the characteristics of the selected representative days, the annual degradation estimation accuracy, and the economic implications. Finally, Section IV concludes the paper and summarises the main findings.

II. METHODOLOGY

Figure 1 shows the high-level workflow used in this study. First, the annual load, solar, and wind time-series data are applied to compute the normalised P_{net} . Then, five representative days are selected, weighted, and used in the Proposed-II degradation analysis, and the final results are compared with the Base Case. In the Proposed-II framework, each representative day is simulated separately and then repeated to estimate the long-term daily degradation behaviour in a computationally efficient way. The weighted degradation contributions of all representative days are then combined to estimate the annual battery degradation. By contrast, the Base Case uses the full 365-day dataset and performs a day-by-day simulation over multiple years until the battery reaches EoL. Therefore, the Base Case gives the most accurate benchmark, while Proposed-II provides a much faster alternative for estimating degradation [6].

A. Degradation-aware representative-day selection

A key issue in representative day analysis is how to select the most appropriate days to capture battery degradation behaviour. A common approach is to first group similar daily P_{net} profiles into clusters and then choose the average profile of each cluster as the representative day, as in our previous study in [6]. Another solution is the most drought day, where solar and wind generations are very low and the net demand is high. At first, such a day may appear to be the worst case for the battery. However, this is not always true from a degradation point of view.

Battery degradation is strongly influenced by battery activity, especially repeated charge and discharge. On a very severe drought day, the battery may experience a long discharge period, but the charging opportunity can be very limited. In such a case, the battery may only see one discharge event, or at most a half cycle, because there is little chance to recharge. In contrast, a day with both charging and discharging opportunities can produce more cycles and higher energy throughput within the same day. Therefore, a drought day may not represent the battery behaviour that causes the higher degradation.

For this reason, selecting representative days only from the cluster mean or from the most drought conditions may not be sufficient to accurately estimate degradation. In this work, representative days are instead selected based on degradation-aware characteristics of P_{net} . Two criteria are considered. The first method selects days with strong and relatively balanced positive and negative P_{net} areas, so that both charge and discharge opportunities are present within the same day. The second method selects days with a high number of zero crossings in P_{net} , indicating frequent switching between positive and negative net demand. A high number of zero crossings means that the battery operating condition changes more frequently between charge and discharge. This is important because battery degradation is not determined only by the total charge and discharge energy throughput, but also by how the cycling events are distributed over time. In particular, frequent switching between charging and discharging can indicate a more demanding cycling pattern, which should be considered in degradation estimation [15], [16]. Therefore, the number of zero crossings is used here as an indicator of more intensive and repeated cycling behaviour during the day.

The accuracy of each method is evaluated by comparing its degradation estimation error with the Base Case of [6]. The two proposed methods based on P_{net} are expected to better capture battery activity and improve degradation estimation in the Proposed-II framework of [6] compared to the benchmark of the Base Case.

Equation (1) defines $P_{net}(t)$ as the difference between the load demand (P_{load}) and the renewable energy available from solar ($P_{solar}(t)$) and wind ($P_{wind}(t)$). This net demand fluctuates, becoming positive when the load is higher than the renewable output, and turning negative when renewable generation exceeds the load.

$$P_{net}(t) = P_{load}(t) - P_{solar}(t) - P_{wind}(t), \quad \forall t \quad (1)$$

B. Benchmark representative-day method

A full year, day-by-day simulation with a five-minute timestep gives the most accurate battery degradation result, but it is very slow and difficult to use in optimal battery or hybrid system planning and sizing studies. To reduce this computational burden, representative-day methods are often used. In our previous work [6], the annual dataset of 365 daily profiles was grouped into a small number of clusters using the K-means algorithm. Each cluster contained days with similar net demand behaviour, and one typical day was selected to represent that cluster. The weight of each representative day was equal to the number of original days that belong to its cluster.

This benchmark method can greatly reduce the simulation time while still preserving the main operating patterns of the full-year dataset. For this reason, the previous study in [6] employed K-means-derived representative days as a practical compromise between computational efficiency and modelling accuracy. However, the method mainly focuses on the general similarity of the profile and does not directly consider the degradation behaviour of the battery. In particular, a day close to the cluster mean may not capture the charge-discharge activity that is most important for estimating battery ageing. Accordingly, this study adopts the K-means representative-day method as a benchmark and compares it with the proposed P_{net} -based degradation-aware representative-day selection approaches.

C. Representative-day selection Method-I: Balanced positive/negative area with significant energy throughput

The first proposed method selects representative days based on the balance between the positive and negative areas of the net load profile, P_{net} . The main idea is that a day with both a considerable positive area and a considerable negative area is more likely to create meaningful battery activity. In such a case, the battery has the opportunity to both discharge during deficit periods and charge during surplus periods. This type of behaviour is important for degradation studies because battery ageing is strongly affected by higher energy throughput (or cycling) rather than only by the average net load level.

For each day d , the positive area of the net load profile is calculated as

$$A_d^+ = \sum_t \max(P_{net,d}(t), 0) \quad (2)$$

where A_d^+ represents the total positive area of the daily net load curve in Eq. 2. This corresponds to periods when the load demand is higher than the available renewable generation, indicating a potential battery discharge requirement.

Similarly, the negative area is calculated as

$$A_d^- = \sum_t \max(-P_{net,d}(t), 0) \quad (3)$$

where A_d^- represents the total negative area of the daily net load curve in Eq. 3. This corresponds to periods when

renewable generation exceeds the load demand, indicating a potential battery charging opportunity.

To identify days that contain strong charge and discharge activity, a score is assigned to each day as

$$Score_d = A_d^+ \times A_d^- \quad (4)$$

A high score in Eq. (4) is obtained only when both A_d^+ and A_d^- are large. Therefore, this method naturally favours days that contain substantial positive and negative net load areas within the same day. By contrast, a day with only a large positive area or only a large negative area will receive a lower score because it does not reflect strong bidirectional battery activity.

Based on this criterion, the days with the highest scores are selected as representative days. These days are expected to better preserve the battery-relevant operational behaviour of the full-year dataset and, therefore, improve the accuracy of degradation estimation within the proposed framework.

D. Representative-day selection Method-II: Zero-crossing based selection

The second proposed method selects representative days based on the number of zero crossings in the daily net load profile, P_{net} . The main idea is that whenever the net load changes sign, the battery operation also changes. A positive value of P_{net} means that the load is higher than the available renewable generation, so the battery is more likely to discharge. In contrast, a negative value of P_{net} means that renewable generation is greater than the load, so the battery is more likely to charge. Therefore, frequent sign changes in P_{net} indicate repeated switching between charging and discharging conditions within the same day.

For each day, the number of zero crossings is counted over the time horizon. To avoid counting very small fluctuations caused by noise or near-zero values, a symmetrical deadband is applied on the net demand before detecting the sign change. In this way, only meaningful changes in operating direction are considered.

A day with a high number of zero crossings is expected to produce more alternating battery activity, which is important for degradation analysis. This is because repeated switching between charging and discharging can better represent the cycling behaviour that contributes to battery ageing [15], [16]. However, for two days with the same $Score_d$, the day with more frequent switching between charging and discharging is expected to cause higher battery degradation. Therefore, zero crossings help distinguish between days with similar overall activity but different cycling intensity.

Based on this criterion, the days with the highest zero-crossing counts are selected as representative days. These days are expected to better preserve the temporal variability and charge-discharge alternation that are relevant for battery degradation studies. As a result, this method provides a degradation-aware alternative to conventional representative-day selection approaches that mainly focus on overall profile similarity.

E. Weight assignment

After selecting the representative days, the next step is to determine how many original days each representative day should stand for in the annual dataset. This is done by assigning every original day to the most similar representative day based on the daily net load profile, P_{net} . For each original day d , its full daily profile, P_{net} , is compared with the profile of each selected representative day. The similarity is measured using the Euclidean distance. For an original day d and a representative day r , the distance is calculated as

$$Dist_{d,r} = \sqrt{\sum_t (P_{net,d}(t) - P_{net,r}(t))^2} \quad (5)$$

where, in Eq. 5, $P_{net,d}(t)$ is the net load value of day d at time step t , and $P_{net,r}(t)$ is the corresponding net load value of representative day r at the same time step.

Each original day is then assigned to the representative day with the minimum Euclidean distance, that is,

$$r^*(d) = \arg \min_r Dist_{d,r} \quad (6)$$

where, in Eq. 6, $r^*(d)$ denotes the representative day that is closest to the original day d . Once all original days are assigned, the weight of each representative day is calculated as the total number of days assigned to it. Therefore, the weight of representative day r can be written as

$$W_r = \sum_{d=1}^{365} \mathbf{1}(r^*(d) = r) \quad (7)$$

where, in Eq. 7, $\mathbf{1}(\cdot)$ is an indicator function that takes the value 1 when the condition is true and 0 otherwise. In this way, each representative day receives a weight that reflects how many days in the full year it represents. The sum of all representative-day weights is therefore equal to the total number of days in the year, i.e.,

$$\sum_{r=1}^{N_{rep}} W_r = 365 \quad (8)$$

where, in Eq. 8, N_{rep} is the number of selected representative days. This weighting process ensures that the reduced representative-day dataset preserves the overall contribution of all original days throughout the year. As a result, the degradation estimated from the representative days can be scaled properly to obtain the annual degradation result.

F. Embedding in Proposed-II degradation framework

The proposed representative-day selection methods are incorporated into the Proposed-II degradation framework and case study developed in our previous work [6]. In that framework, representative days are used to estimate battery degradation in a computationally efficient way, without performing a full day-by-day simulation over many years until the battery reaches its end of life. Instead of selecting representative days only based on overall profile similarity, the present study improves this step by selecting days that are more relevant to battery activity and degradation behaviour.

After the representative days are selected and their weights are assigned, each representative day is individually simulated using the battery operation model. This provides the daily battery operating trajectories, such as the state of energy and the corresponding state of charge profile. As in the earlier Proposed-II framework, the simulation is carried out using the battery capacity trajectory of the final operational year, so that the battery behaviour is evaluated under a more realistic degraded-capacity condition.

The Proposed-II framework used in this paper is the same as in our previous study [6], except for the way the representative days are selected. In Proposed-II, battery degradation is not estimated from only one occurrence of each representative day. Instead, each representative day is repeated for an extended number of days so that the degradation trend moves beyond the initial high-degradation stage and reaches the more stable linear region, which provides a more reliable basis for estimating annual degradation. In our earlier work, the representative days were selected mainly from cluster-based typical profiles, whereas in this study they are selected using P_{net} -based degradation-aware criteria. In this way, the framework keeps the computational advantage of Proposed-II while improving the likelihood that the selected days better capture the charge-discharge behaviour that drives battery ageing.

For each representative day d , the repeated state-of-charge and temperature profiles are passed to the BLAST degradation model [17] to calculate the degradation caused by that day type. This gives the daily degradation value, denoted by $e_{loss,d}$. The annual degradation contribution of each representative day is then obtained by multiplying this daily degradation value by its assigned weight W_d . Therefore, the representative-year degradation can be written as

$$e_{loss,yr}^{rep} = \sum_{d=1}^{N_{rep}} W_d e_{loss,d} \quad (9)$$

Once the annual degradation is obtained by Eq. 9, the battery life can be analytically estimated by dividing the allowable capacity loss by the representative annual degradation. In this way, the Proposed-II framework avoids a full multiyear simulation until EoL, while still preserving the degradation effect of battery-relevant daily operation patterns.

G. Performance metrics

Two metrics are used to compare the representative-day selection methods against the Base Case.

The first metric is the annual degradation estimation error, which shows how closely each method predicts the yearly battery degradation:

$$Err^{AD} = \left| \frac{AD_m - AD_{base}}{AD_{base}} \right| \times 100 \quad (10)$$

where, in Eq. 10, AD_m is the annual degradation estimated by the method m and AD_{base} is the annual degradation of the Base Case.

The second metric is the net present cost (NPC) estimation error (Err^{NPC}), which measures the economic impact of the degradation estimation error:

$$Err^{NPC} = \left| \frac{NPC_m - NPC_{base}}{NPC_{base}} \right| \times 100 \quad (11)$$

where, in Eq. (11), NPC_m and NPC_{base} are the net present costs of the method m and the Base Case, respectively. The NPC calculation methodology has also been explained in detailed in our previous study [6].

H. Method comparison

In this study, three representative-day selection methods are compared. The first is the benchmark K-means method used in our previous work, where similar daily P_{net} profiles are grouped into clusters and a typical day is selected from each cluster. The second is the proposed area balance method, which selects days with high positive and negative P_{net} areas on the same day. The third is the proposed zero-crossing method, which selects days with frequent sign changes in P_{net} . For a fair comparison, all methods use the same number of representative days and the same weighting process as given in Table I. Therefore, the main difference lies only in how the representative days are selected.

TABLE I: Comparison of representative-day selection methods

Method	Selection basis	Days	Weighting
K-means	Cluster-based typical day	5	Euclidean distance
Area balance	Highest $A_d^+ \times A_d^-$	5	Euclidean distance
Zero crossing	Highest sign-change count	5	Euclidean distance

III. RESULTS AND DISCUSSION

A. Characteristics of representative days selection methods

Figures 2a and 2b show the five representative days chosen by the two selection methods for the data for 2023. In Method-I, the selected days are chosen based on a balanced positive and negative P_{net} area. The selected days are Day 325, Day 186, Day 6, Day 221, and Day 202, with corresponding weights of 29, 14, 82, 31, and 209 days, respectively. These plots show that the selected days contain strong charging and discharging opportunities within the same day, although the balance is not exactly the same for all days. For example, some days show more surplus-dominant behaviour, while others show stronger deficit periods. In Method-II, the selected days are chosen based on frequent zero crossings of P_{net} , which means that the net demand changes sign many times during the day. The selected days are Day 270, Day 128, Day 271, Day 148, and Day 125, with corresponding zero-crossing counts of 21, 17, 13, 13, and 12, and associated weights of 73, 49, 48, 31, and 164 days, respectively. Compared to Method-I, these representative days better capture repeated switching between positive and negative P_{net} , which is more closely related to the frequent battery charge-discharge activity. The results also show that the annual dataset is still dominated by some repeated patterns, as some representative days receive

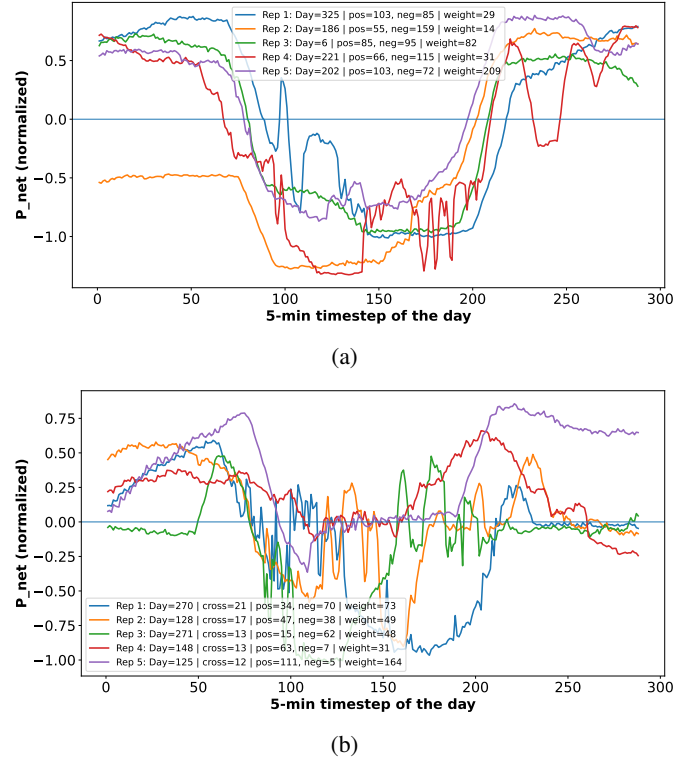


Fig. 2: Proposed representative-day selection for year 2023; (a) Method-I: Balanced positive/negative area, while (b) Method-II: Zero-crossing based selection.

much larger weights than others. The other years (2014 - 2022) selection results are shown in the Appendix [18].

B. Degradation estimation accuracy

Table II compares the annual battery degradation estimated by the different representative-day selection methods. The Base Case and the mean representative-day selection results are taken from our previous study in [6]. The two new methods, namely Selection Method-I (balanced positive and negative area) and Selection Method-II (crossing-zero), are proposed in this work and tested within the same Proposed-II framework of our previous work.

Both new representative-day selection methods give results much closer to the Base Case than the cluster-mean selection benchmark, as illustrated in Fig. 3. Based on the average relative error over the 10 studied years (2014–2023), the cluster-mean selection method gives an error of 21.83%, while Selection Method-I and Selection Method-II reduce this error to 5.12% and 6.71%, respectively. This clearly shows that choosing representative days based on degradation-relevant P_{net} can improve the degradation estimation accuracy.

Among the two new methods, Selection Method-I provides the best overall performance. Although Selection Method-II also significantly improves results compared to the cluster-mean selection method of [6], its overall accuracy is slightly lower than Selection Method-I. Therefore, the results confirm that the proposed degradation-aware representative-day selec-

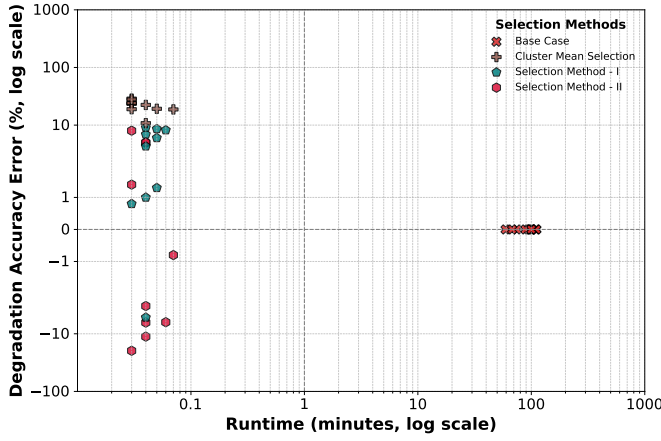


Fig. 3: Comparison of battery degradation estimation accuracy and runtime across different representative day selection methods in ten years of simulation.

tion methods improve the accuracy of Proposed-II framework of our existing study, with Selection Method-I giving the best overall agreement with the Base Case within similar existing simulation runtimes.

C. Economic implications

The EoL of the battery in Table III is calculated from the annual degradation of the battery reported in Table II. In this study, it is assumed that the battery reaches EoL when its state of health (SoH), or maximum capacity, decreases to 80%. The cost models from [6] have been applied to all years (2014-2023) to determine the battery NPC over a 30-year project lifetime. Battery degradation significantly impacts the number of replacements required within the project's duration, as even minor differences in annual degradation rates can lead to notable variations in battery replacement schedules, associated replacement costs, and salvage values at the project's conclusion. In contrast, other cost components, such as initial capital investment and operational expenditures, remain consistent in all the approaches evaluated. Table IV summarises the net present cost (NPC) calculated for each method using ten years of data, and the NPC error with respect to the Base Case for each year (illustrated in Figure 4). The results clearly show that Selection Method-I achieves the lowest NPC error in all evaluated years, ranging from only -3% to 4%. This consistently high level of accuracy underscores the

TABLE II: Annual battery degradation estimated by different representative-day selection methods (*Bold value in each row shows the one closest to the Base Case*).

Year	Base Case (%)	Cluster-mean (%)	Method-I (%)	Method-II (%)
2014	1.878	2.419	2.007	2.029
2015	1.890	2.313	1.914	1.982
2016	1.962	2.434	1.977	1.897
2017	1.905	2.350	2.078	1.931
2018	1.968	2.445	1.988	1.842
2019	1.904	2.435	2.068	2.000
2020	2.094	2.320	1.985	1.681
2021	1.990	2.367	2.075	1.767
2022	1.957	2.322	2.117	1.941
2023	1.957	2.333	2.074	1.834

TABLE III: Battery EoL estimated by different representative-day selection methods (*Bold value in each row shows the one closest to the Base Case*).

Year	Base Case (yr)	Cluster Mean (yr)	Method-I (yr)	Method-II (yr)
2014	10.65	8.27	9.97	9.86
2015	10.58	8.65	10.45	10.09
2016	10.19	8.22	10.12	10.54
2017	10.50	8.51	9.62	10.36
2018	10.16	8.18	10.06	10.86
2019	10.50	8.21	9.67	10.00
2020	9.55	8.62	10.08	11.90
2021	10.05	8.45	9.64	11.32
2022	10.22	8.61	9.45	10.30
2023	10.22	8.57	9.64	10.91

TABLE IV: Project NPC based on battery EoL of different representative-day selection methods (*Bold value in each row shows the one closest to the Base Case*).

Year	Base Case (M\$)	Cluster Mean (M\$)	Method-I (M\$)	Method-II (M\$)
2014	11.831	13.015	12.060	12.139
2015	11.843	12.843	11.869	11.934
2016	11.917	13.035	11.929	11.852
2017	11.860	12.907	12.303	11.886
2018	11.922	13.047	11.940	11.790
2019	11.858	13.035	12.270	12.771
2020	12.335	12.857	11.936	11.569
2021	11.941	12.934	12.289	11.696
2022	11.911	12.862	12.411	11.897
2023	11.911	12.880	12.289	11.780

practical viability of the Selection Method-I in cost-sensitive planning studies.

IV. CONCLUSIONS

This paper showed that conventional representative-day selection methods are not always suitable for battery degradation studies. In particular, a drought day is not necessarily the worst day for degradation analysis because low renewable generation does not always lead to high battery activity. The results showed that representative days selected using battery activity characteristics based on P_{net} can improve the accuracy of estimating the degradation of the existing Proposed-II framework.

Among the tested methods, the balanced positive and negative P_{net} -area method gave the best overall performance, while the zero-crossing-based method also improved the results compared to the benchmark mean selection method.

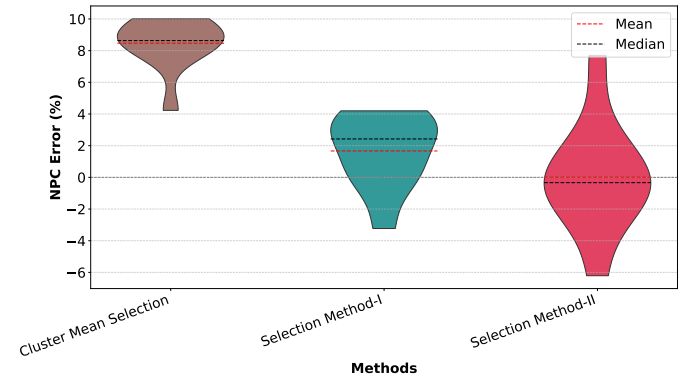


Fig. 4: NPC error distribution by each selection methods compared to the Base Case.

These findings confirm that representative-day selection should consider battery-relevant net-demand behaviour with more care in degradation-focused studies. In future work, the proposed methods can be integrated into a bi-level battery sizing framework to size the battery using more accurate degradation estimates, which can help avoid oversizing or undersizing under different load and renewable conditions. Future studies may also investigate hybrid scoring methods, an adaptive number of representative days, and direct degradation-aware clustering for representative-day selection.

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